
SalinBloc™

figures/salinbloc-skid.png

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CHAPTER

About this manual

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1.1 Purpose and scope

This manual describes the SalinBloc™ container: a modular SonixED™ electro dialysis unit for water desalination and brine valorization. The standard product is a single container fitted with the stack, dosing and control equipment; an optional second container adds pretreatment ahead of it for sites whose raw water needs filtering first (chapters 3 and 8). Part I of this manual covers the unit as a whole (safety, installation, commissioning, day-to-day operation). Part II gives one self-contained chapter per compartment inside the container — pumps, stacks, valves, dosing, sensors, control cabinet, the diluate, concentrate, CIP and electrolyte tanks (each its own chapter, since they hold very different streams — and, for the CIP tank, very different chemicals at different times — under very different conditions), and telemetry — each with its own specifications, operating notes, maintenance and troubleshooting, so it can be consulted (or printed/exported) on its own. Part III closes with shutdown/storage and reference appendices.

SonixED™ is HydroVolta NV's electro dialysis-reversal (EDR) technology, combined with an integrated ultrasound antifouling stage and a custom ion-selective membrane coating. Hydrovolta NV (Leuven, Belgium, founded 2016) developed SonixED™ over eight years of lab validation with research partners including VITO, and holds a granted European patent (2021) on the combined EDR + ultrasound process. SonixED™/SalinBloc™ units are in active use or pilot deployment at sites in Belgium, Morocco, Australia (Perth), Brazil and Saudi Arabia; see section 3.2 for field performance data published by HydroVolta.



NOTICE

Several sections in this manual are marked **TODO**. These mark information that was not found in HydroVolta's engineering documentation or public materials at the time of writing (July–August 2026) and must be supplied by HydroVolta engineering, or confirmed by inspection/measurement on an actual unit, before this manual is issued to a customer. Every other statement in this manual is grounded in a specific HydroVolta engineering document, datasheet, or the hydrovolta.com website — but the underlying documents mostly describe pilot/demonstration containers (Farys, Ganzepoot, BASF, Morocco, and others), not a finalized, branded “SalinBloc™” bill of materials. Treat pilot-specific facts as a well-informed engineering basis for the standard product, not as a guaranteed as-shipped specification, until confirmed against a specific unit's documentation. Notably, the **PLC control program itself is out of scope for this edition** of the manual — it is still under development by HydroVolta's automation team, and the relevant sections (HMI screen navigation, exact control logic, full setpoint list) will be completed once that work is finished.

1.2 Who this manual is for

This manual is written for the people who take delivery of, install, commission, operate and maintain a SalinBloc™ unit: site engineers, installation contractors, plant operators, and HydroVolta service technicians. Readers are assumed to have basic familiarity with industrial piping, low-voltage electrical systems and PLC-controlled equipment; no prior knowledge of electrodialysis is assumed (see the Glossary, appendix D, for unfamiliar terms).

1.3 Symbols and signal words

Hazard and information boxes in this manual follow the signal-word convention below, aligned with ISO 3864 / ANSI Z535 safety-color practice.



DANGER

Indicates a hazard that, if not avoided, **will** result in death or serious injury.



WARNING

Indicates a hazard that, if not avoided, **could** result in death or serious injury.



CAUTION

Indicates a hazard that, if not avoided, could result in minor or moderate injury.



NOTICE

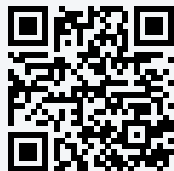
Indicates a risk of property damage or equipment malfunction, with no personal-injury hazard.

NOTE

Highlights additional information that helps the reader but is not itself a hazard warning.

1.4 Online documentation and QR code

A QR code on the container's control cabinet links to the latest online version of this manual and to the unit's factory quality report.



Scan to open the online manual and this unit's quality report at <https://hydrovolta.com/salinbloc-manual>.

1.5 Contact and support

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Web: hydrovolta.com. See section 7.6 for when to contact service.

CHAPTER

Safety

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WARNING

Read this entire chapter before installing, operating or servicing the SalinBloc™ unit. No unit-level hazard analysis, PPE schedule or lockout/tagout procedure specific to SalinBloc™/SonixED™ exists yet in HydroVolta's engineering library — only generic contractor-safety material (tool safety sheets, an electrical risk assessment template, a VCA contractor-competency handbook) and the Safety Data Sheets for the dosing/CIP chemicals. The hazards, PPE and isolation guidance below are derived from those chemical SDSs and from the electrical/pressure/DC figures documented elsewhere in this manual (chapters 10 and 14); they are a defensible starting point, **not** a substitute for a formal hazard analysis, which HydroVolta engineering/EHS must still produce and sign off before this manual is issued to a customer.

2.1 Intended use

The SalinBloc™ unit is intended for the electrodialysis desalination or ion reduction of brackish groundwater, industrial process water, or brine, in the total dissolved solids range HydroVolta publishes as the technology's operating envelope — approximately 1,500–20,000 ppm TDS (see section 3.2) — and within the feedwater quality limits stated for the pretreatment and stack stages (chapters 8 and 10). It is not intended for seawater desalination (HydroVolta states seawater, >35,000 ppm, as outside SonixED™'s primary use case) and is **not** currently certified as a drinking-water treatment system: the 03_UK DWI reg 31 approval folder in HydroVolta's datasheet library contains only generic third-party guidance on how Reg 31 approval works (not an actual approval or application), and the 04_NSF folder contains only a general NSF/ANSI 60/61/372 adoption survey (not an NSF listing) — confirmed by direct inspection of both folders. Do not make a potable-water regulatory claim to a customer until formal approval is obtained — see appendix C. Foreseeable misuse includes: feeding water outside the documented turbidity/iron/chlorine/pH envelope (chapter 10) without adequate pretreatment; bypassing or defeating interlocks, alarms or the emergency stop; operating with the cabinet doors open; and mixing CIP chemicals outside the sequence in chapter 12 (in particular, never combine NaOH with feed/process water — see the warning in chapter 12).

2.2 Qualified personnel

Work on the control cabinet, VFDs and any live electrical circuit, including the SonixED™ stack's DC bus (up to 600 VDC, chapter 10), must only be performed by an electrically qualified person as defined in IEC 60204-1. Chemical dosing tank refilling and CIP chemical handling must only be performed by personnel trained on the relevant Safety Data Sheet for that chemical (section 2.4). HydroVolta's own engineering library uses the Belgian/Dutch **VCA** (Veiligheid, Gezondheid en Milieu Checklist Aannemers) contractor-safety certification as its baseline competency reference for site work; treat VCA-certified (or locally equivalent) status as the expected minimum for any contractor performing installation or maintenance work on the unit. General operators (start/stop, HMI monitoring, routine visual checks) do not need electrical or chemical-handling qualifications but must be trained on this manual and on the local isolation procedure (section 2.7) before working unsupervised.

2.3 Personal protective equipment

- **Chemical dosing / CIP work** (antiscalant, HCl, NaOH, NaOCl, H₂SO₄ — all corrosive per their SDS, section 2.4): chemical-resistant gloves, safety glasses/goggles with a face shield for tank filling or CIP setup, and a chemical-resistant apron. Have an eyewash/emergency shower available wherever CIP or dosing-tank work is performed.
- **Control cabinet / electrical work**: PPE appropriate to the task and the incoming supply voltage (3×400 V AC, chapter 14) per the site's local electrical safety regulation and the qualified person's own risk assessment; always verify isolation (section 2.7) before opening the cabinet.
- **General site / mechanical work** (pumps, valves, piping): safety shoes and, where local site rules require it, hi-vis clothing and hearing protection near running pumps.

2.4 Main hazards

- **Electrical:** the incoming 3×400 V AC cabinet supply (chapter 14), and — specific to this technology — the SonixED™ stack's own DC bus, which is energized to **60–600 V DC at 3–20 A** during normal operation (chapter 10). This DC bus is a hazard independent of the AC supply and must be isolated and verified dead separately before any stack or manifold work.
- **Chemical:** the antiscalant, HCl, NaOH, NaOCl and H₂SO₄ used for dosing and CIP (chapter 12) are corrosive per their Safety Data Sheets (Carl Roth SDS on file in HydroVolta's engineering library). **Never mix NaOH with feed or process water** — HydroVolta's own CIP procedure documents this as a scaling hazard, and mixing acid and hypochlorite streams carries a toxic-gas risk common to all such systems; keep chemical dosing lines and procedures strictly separated as described in chapter 12.
- **Pressurized piping:** the hydraulic circuit operates up to 2.5 bar (stack maximum 3 bar, chapter 10) and the pretreatment UF stage's CEB/backwash cycle briefly reaches similar pressures (chapter 8); depressurize before opening any fitting.
- **Moving parts:** process pumps, dosing pumps, and automated valve actuators (chapters 9 and 11) start automatically under PLC control without warning — treat every pump and valve as capable of starting at any time unless the unit is isolated.

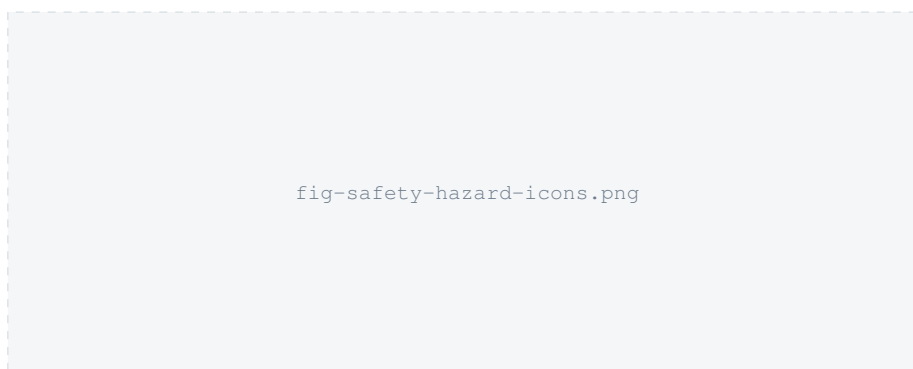


Figure 2.1. Suggested figure: a labeled diagram of the container's exterior and interior showing each hazard zone above (electrical cabinet / DC stack bus, chemical dosing area, pressurized piping, pump/valve zones) with its corresponding ISO 7010 warning pictogram.

2.5 First aid and chemical spill response

The following is transcribed directly from the manufacturer Safety Data Sheets on file for the dosing/CIP chemicals (Carl Roth SDS, Section 4 “First aid measures” and Section 6 “Accidental release measures”; text is consistent across NaOH, HCl, NaOCl and H₂SO₄). Post this section, or an equivalent quick-reference card, wherever chemicals are handled.



Eye contact: flush immediately with plenty of flowing water for **10 to 15 minutes**, holding eyelids apart, and consult an ophthalmologist. Protect the uninjured eye. **Skin contact:** take off all contaminated clothing immediately; wash the affected skin immediately with plenty of water (for H₂SO₄: first mechanically remove the chemical, e.g. dab away with wadding, then wash with a mild cleansing agent and water); seek immediate medical treatment — corrosive injuries left untreated are hard to cure. **Inhalation:** move to fresh air; seek medical advice if symptoms persist or in any case of doubt. **Ingestion:** rinse the mouth immediately and drink plenty of water; call a physician immediately — risk of gastric/esophageal perforation from the corrosive effect.

Spill response (non-emergency personnel only; do not attempt without the PPE in section 2.3): put on PPE, avoid contact with skin, eyes and clothing, and do not breathe vapour/spray. Cover nearby drains to protect surface/ground water — for HCl and H₂SO₄ specifically, the product must be neutralized before any discharge to a drain or sewage plant. Absorb the spill with a liquid-binding material (sand, diatomaceous earth, or an acid/universal binding agent), then place the absorbed material in an appropriate container for disposal. Ventilate the affected area, especially after an HCl release.

2.6 Safety devices and emergency stop

HydroVolta's HMI design specification (chapter 6) confirms a **hardware Emergency Stop button that bypasses the PLC entirely** — see section 6.9 for the full recovery procedure. TODO: its physical location on the control panel/cabinet, and whether more than one E-stop station exists (e.g. one per container, or one per access point), was not confirmed in the documents reviewed — verify against the as-built electrical schematic and IEC 60204-1 on the actual unit before publication. No document describing pressure-relief devices or leak detection was found.

2.7 Isolating the unit before work

No HydroVolta-authored lockout/tagout procedure was found; until one exists, apply this general sequence before any maintenance, sensor, or piping work:

1. Stop the process via the normal `Stop` sequence (chapter 6).
2. Isolate and lock out the incoming electrical supply at the main breaker; verify absence of voltage at the cabinet terminals before touching any wiring.
3. Separately verify the SonixED™ stack's DC bus is de-energized — it operates independently of the AC supply isolation point and can present up to 600 V DC (chapter 10); never assume it is safe just because the AC breaker is open.
4. Isolate and depressurize the hydraulic circuit (close feed/product/ drain isolation points, section 4.4) before opening any fitting.
5. Where chemical dosing lines will be disturbed, close the relevant tank isolation valve and follow the PPE guidance in section 2.3.

TODO: this sequence is assembled from the general electrical/hydraulic/DC facts documented elsewhere in this manual, not from a validated HydroVolta LOTO procedure — HydroVolta engineering/EHS must review and formalize it, including lock/tag hardware and a permit-to-work step if required by the customer's site rules.

2.8 Safety labels on the container

A folder named `17_Photos/Container stickers` exists in HydroVolta’s engineering library, but on inspection it contains only **brand artwork** (the Hydro-Blue and Volta-Yellow logo marks, the HydroVolta wordmark, a “water desalination” decal, and a door logo) — **no hazard/warning pictogram stickers were found there or anywhere else in the library**. This is a real gap: before a unit ships, it needs a proper ISO 7010 hazard-label set (electrical, DC stack, chemical, pressure — matching section 2.4) actually affixed and photographed, and this section rewritten from those photographs.



Figure 2.2. Suggested figure: photograph(s) of the container exterior and control cabinet with each safety/identification label numbered, plus a key table listing what each label means and where it is mounted — to be produced once a real hazard-label set has been designed and affixed to a unit (none currently exists in HydroVolta’s documentation).

System overview

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3.1 What the unit does

The SalinBloc™ unit is built around HydroVolta's SonixED™ technology, which combines three mechanisms in one automated platform:

1. **Electrodialysis reversal (EDR):** an applied DC electric field draws dissolved ions — not water — through ion-exchange membranes from the diluate stream into the concentrate stream. Because water itself does not have to be forced through a membrane (unlike reverse osmosis), the process runs at much lower hydraulic pressure. Periodically reversing polarity cleans the membranes in place without chemicals and gives more stable performance as feed salinity varies.
2. **Integrated ultrasound antifouling:** ultrasound transducers built into the stack create acoustic cavitation at the membrane surface, disrupting scale nucleation and helping prevent silica fouling — this is the element of SonixED™ that HydroVolta holds a granted European patent on (2021).
3. **Custom membrane coating:** a proprietary coating gives the membranes monovalent-ion selectivity, so the process can target nitrate, sodium, chloride and fluoride removal while retaining calcium and magnesium in the product water.

The result is a saline feed stream split into a low-salinity product stream (*diluate*) and a concentrated reject stream (*concentrate/brine*), with chemical-free ultrasound antifouling keeping the membrane stack clean and extending electrode/membrane life (chapter 10). Typical applications: groundwater desalination and selective ion (e.g. nitrate) removal, industrial process-water reuse, brine valorization/concentration, and Zero Liquid Discharge (ZLD) support.



Figure 3.1. Exterior view of a delivered SalinBloc™ container on site.

3.2 Field performance reference

HydroVolta publishes the following performance comparison from a 90-day industrial pilot (February–May 2023, cooling-tower blowdown at a chemical industry facility in Germany, feed TDS $\approx$ 4,500 ppm, silica 85 mg/l, high hardness), run head-to-head against reverse osmosis (RO) and a standard EDR system on the same feedwater (source: hydrovolta.com/technology.html and [/projects.html](https://hydrovolta.com/projects.html)). These figures describe that specific pilot and HydroVolta's general SonixED™ technology, not a performance guarantee for any individual SalinBloc™ unit; use them as context for what the technology is designed to achieve, and always verify against the site-specific commissioning data recorded in appendix A.

Table 3.1. Germany 2023 industrial pilot — SonixED™ vs. RO vs. standard EDR (source: hydrovolta.com)

Parameter	SonixED™	RO	Standard EDR
Water recovery	90–98%	50–75%	70–85%
Energy consumption	0.19– 0.7 kWh/m ³	2.0– 4.0 kWh/m ³	1.0– 2.0 kWh/m ³
Operating pressure	0.5–2.5 bar	25–35 bar	1–2 bar
Chemical dosing	Minimal, no antiscalant	Required	Antiscalant needed
CIP frequency	0.6×/month	1×/month	1×/month
Total treatment cost	\$0.30– 0.40/m ³	\$0.60– 1.00/m ³	\$0.40–0.80/m ³

HydroVolta states SonixED™'s general operating envelope as feed TDS 1,500–20,000 ppm (“strong fit” also for high silica >30 mg/l and high hardness >300 mg/l Ca+Mg feeds); seawater (>35,000 ppm) is stated as not the primary use case. Active and piloted SonixED™/SalinBloc™ deployments referenced by HydroVolta include a flagship groundwater desalination project in Morocco (seawater-intruded coastal aquifer, $\approx$ 7,000 ppm feed), a multi-site utility pilot with Water Corporation in Perth, Australia (selective nitrate and salinity removal), the Germany industrial pilot above, and earlier-stage projects in Brazil and Saudi Arabia.

3.3 Container layout

The **standard SalinBloc™ unit is a single container** (20 ft or 40 ft, High-Cube or Standard) housing the SonixED™ skid(s) (chapters 9 to 11 and 14) and everything downstream of the feed connection — it does **not** include a pretreatment stage, so feed water must already meet the stack's feedwater envelope (chapter 10) before it reaches the unit. **Pretreatment (chapter 8) is an optional add-on**, not part of the standard product: where a site's raw water needs filtration first, HydroVolta can supply a *second, separate pretreatment container*, connected hydraulically ahead of the standard unit. Everything below describes the standard single-container unit unless a step is explicitly marked as part of the optional pretreatment container. A documented 20 ft general-arrangement layout (drawing 2308CA1) shows internal dimensions of 6058 × 2438 × 2896 mm, housing a bank of SonixED™ stack racks, an electrical/control cabinet, a dosing/chemical section, three overhead storage tanks, and interconnecting piping; a companion top-view drawing (2308CA2) lays the skids out along the container floor: chemical skid, pump skid, tank skid, control 1 skid, filter skid, SonixED™ skid, control 2 skid.

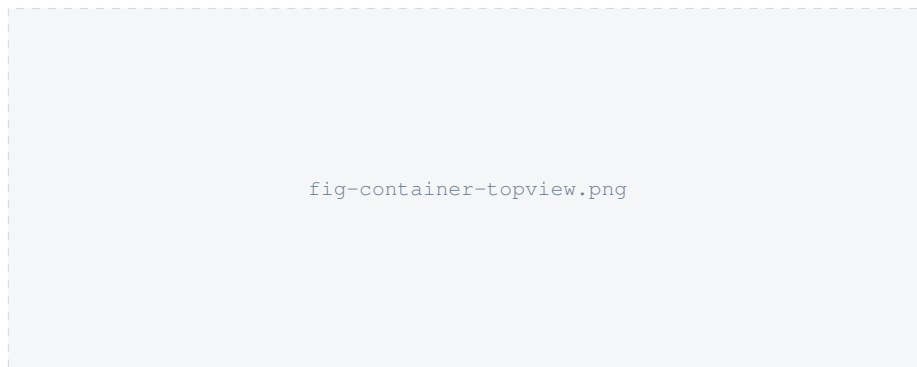


Figure 3.2. Suggested figure: the skid top-view layout (based on drawing 2308CA2), redrawn/simplified for this manual, with each skid numbered and cross-referenced to its Part II chapter.

3.4 Process flow

Standard unit (no pretreatment):

1. Feed water enters the container directly (2" connection) — it must already meet the stack's feedwater envelope (chapter 10); the unit does not filter or otherwise condition it first.
2. Feed water is split and pumped through parallel SonixED™ stacks (chapters 9 and 10), with three internal streams per stack: **diluate** (becomes the product/permeate), **concentrate** (becomes the brine/reject), and **electrolyte** (a separate loop feeding the electrode compartments, chapter 18).
3. Product (permeate) water exits the container via a 2" outlet connection.
4. Reject/concentrate is routed to a drain tank via a 3" drainage connection.

With the optional pretreatment container added (only relevant if that add-on was ordered, chapter 8): feed water instead enters the separate pretreatment container first (2½" connection from source); disc filtration and ultrafiltration condition it there; filtered water then leaves via a 2" connection into the main unit's feed inlet above, and the pretreatment container's own backwash/ drainage is tee'd together with the main unit's drain line to the shared drain tank.

3.5 Main components

This section is a one-page index of every compartment covered in its own chapter later in this manual (Part II) — use it to jump straight to the part you need.

- **Pretreatment and filtration skid (optional add-on, not part of the standard unit)** — chapter 8
- **Feed and process pumps** — chapter 9
- **SonixED™ stack modules** — chapter 10
- **Valves and manifold** — chapter 11
- **Dosing system** — chapter 12
- **Instrumentation and sensors** — chapter 13
- **Control cabinet, PLC and HMI** — chapter 14
- **Diluate tank** — chapter 15
- **Concentrate tank** — chapter 16
- **CIP tank** — chapter 17
- **Electrolyte tank** — chapter 18
- **Telemetry and remote connectivity** — chapter 19

3.6 Control cabinet and HMI

The control system consists of a PLC with a touchscreen HMI, manual/override controls, and RS485 communication, housed in an IP55-rated cabinet. The skid ordering code allows two control-box configurations — **CB0** (PLC only) and **CB1** (PLC + touchscreen HMI) — and three cloud-connectivity tiers: **CL0** (local only), **CL1** (cloud monitoring), **CL2** (full cloud monitoring and control). See chapter 14 for details.

3.7 Nameplate and identification

TODO: no nameplate drawing, CE-marking template with filled-in fields, or Declaration of Conformity content was found (the 02_DoC's and 05_SonixED_stack CE-marking folders in HydroVolta's engineering library are confirmed empty — see appendix C). The confirmed identification data is the datasheet labeling convention "Model: SONIXED-SKID, Version: 2025-A7" and the ordering code format SONIXED-SKID-A-S-M-PR-V-CB-CL (application type A0/A1/A2, stack count 1/2/3, measuring points 6/9/12, pressure sensors 6/9/12, valve count 2/4/6, control-box option CB0/CB1, cloud option CL0/CL1/CL2). Each skid physically measures approximately $3040 \times 1340 \times 2020$ mm (l × b × h), on an aluminum-profile frame with a PVC/thermoplastic-spacer housing, mounted on lockable wheels for in-container mobility. HydroVolta's public SalinBloc™ brochure states container throughput by format: a **20 ft** unit handles 2–30 m³/h (≈ 44 –660 m³/day at 22 h/day operation); a **40 ft** unit handles 15–60 m³/h (≈ 330 –1,320 m³/day) — see section 3.2 and appendix A. No container weight figure was found in any source reviewed.



Figure 3.3. Suggested figure: photograph of the unit's nameplate once affixed, showing serial number, manufacture date, voltage/power rating, IP rating and CE mark.



CHAPTER

Installation

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4.1 Delivery and unloading

TODO: no document describing crane/forklift unloading, container weight, or lifting-point locations was found. The container general-arrangement drawings (2308CA1/2308CA2, appendix B) likely show lifting points but have not yet been reviewed for this purpose.



Figure 4.1. Suggested figure: photograph or diagram of the container being lifted/unloaded on site, with lifting points marked, once the procedure is confirmed.

4.2 Site requirements

The standard unit is a single ISO shipping container (chapter 3), 20 ft/40 ft, High-Cube or Standard. If the optional pretreatment add-on is included, the site needs space for a **second** container alongside it — in HydroVolta's documented BASF pilot, the pretreatment unit used a 10 ft container. The site must provide a water source (the pilot drew from a cooling-tower basin via a submersible pump) and space for a drain tank. TODO: ambient temperature range, ground/ foundation/leveling requirements and clearances beyond what the pilot layout implies were not found in a dedicated site-requirements document.

4.3 Positioning the container

Where the optional pretreatment container is also fitted: from the piping documented for commissioning (section 4.4), its filtered-water outlet (2") must be within pipe-run distance of the main unit's feed inlet (2"), and both containers' 3" drain outlets should be positioned so they can be tee'd together to a shared drain tank.

4.4 Water connections

Standard unit:

- Feed inlet: 2" (M) — feed water must already meet the stack's feedwater envelope (chapter 10) before connection.
- Product-water outlet: 2" (M).
- Drain outlet: 3" (M).
- Fit O-rings and PTFE tape at every threaded joint.

If the optional pretreatment container is also fitted (chapter 8) — these connections apply only to that add-on, not to the standard unit above:

- Pretreatment feed inlet 2½" (M) from the raw water source (in the pilot, a submersible pump rated 4.5 m³/h at 6 bar); filtered-water outlet 2" (M), connected to the standard unit's feed inlet above; drain outlet 3" (M).
- Pump-to-pretreatment feed line: PVCu, 60 mm, terminated 2½" (F).
- Pretreatment-to-unit line: PVCu, 40 mm.
- Both containers' 3" (M) drain outlets combine via a tee into one line to the shared drain tank.
- Key in-line pretreatment components: an ultrafiltration unit (AKKIM APM 80) via 2" Victaulic couplings, and disc filters (AZUD DF100/DF50/DF20) via 2" (F) 63 mm transition pieces.

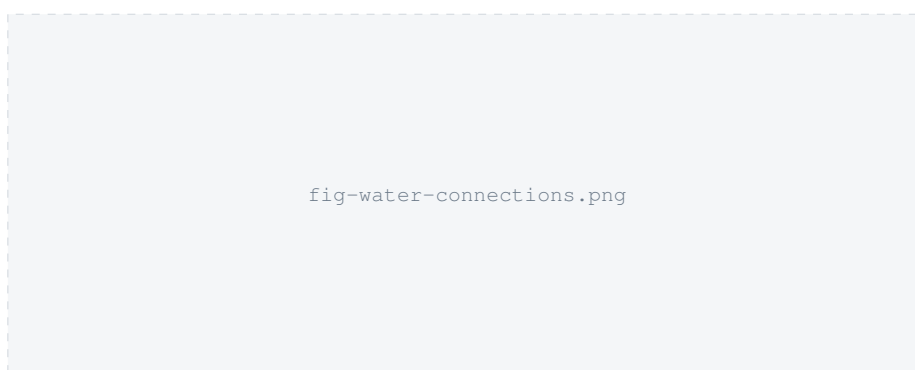


Figure 4.2. Suggested figure: a labeled diagram of the pretreatment/treatment container pair showing every hydraulic connection point (feed, filtered-water, product, drain) with its size and thread type.

4.5 Electrical connection

The control cabinet is IP55-rated. The figures most likely to apply to the **standard unit** come from a separate electrical schematic series for a **treatment container** under the BASF pilot project (folder `2_Electrical/0_Treatment container`, multiple dated AutoCAD Electrical `.dsn` revisions) — TODO: this series was located but not opened (CAD format); a human with AutoCAD Electrical should open the latest revision for the breaker/RCD sizing. The only numeric figures actually extracted so far are from the *pretreatment* container's schematic instead (3×400 V AC, 3P+N+PE per IEC 60445, max. 25 A, 24 V DC control bus for valves/I/O, sheet E1/ document I2) — since pretreatment is an optional add-on (chapters 3 and 8), **do not assume these values apply to the standard single-container unit** until the treatment-container schematic above is actually opened and checked.

4.6 Network and telemetry connection

See chapter 19 for the router, remote-access gateway and cloud dashboard used for network/telemetry connectivity.

4.7 Installation checklist

1. Confirm all hydraulic connections listed in section 4.4 are complete: feed inlet/product outlet/drain (standard unit), plus — only if the optional pretreatment container was ordered — its own feed/outlet/drain and the combined drain-to-tank tee.
2. Confirm electrical supply matches the cabinet's rated voltage/current (chapter 14) before energizing.
3. Confirm network/telemetry connection is established (chapter 19).



CHAPTER

Commissioning

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5.1 Before the first start

Confirm every hydraulic connection in section 4.4 is complete and leak-checked before energizing the cabinet.
TODO: no document describing a formal pre-start electrical isolation/lockout check or inspection checklist was found beyond the piping steps.

5.2 Filling and priming

Standard unit: connect the feed source to the 2" feed inlet and verify the product-water outlet is connected per site requirements. **If the optional pretreatment container is fitted:** install the submersible feed pump in the source water body; connect its outlet to the pretreatment 2½" (F) inlet with 60 mm PVCu pipe; connect the pretreatment filtered-water outlet to the main unit's feed inlet with 40 mm PVCu pipe; join both containers' drain outlets via a tee to the drain tank.

5.3 First power-up

What *does* exist is a real commissioning engineer's log from a pilot startup, which illustrates the general shape of a ramp-up sequence (source: pilot commissioning notes, HydroVolta engineering library) — treat it as an informative example, not a validated procedure for the standard product:

1. Start the feed and process pumps at a low VFD frequency (~ 35 Hz in the logged example) to establish flow gently rather than at full speed.
2. Increase the dosing-pump VFD frequency in steps (e.g. to ~ 60 Hz) once feed flow is confirmed stable.
3. Confirm each stream reaches a stable pressure and flow reading before proceeding — the log explicitly checks “stable pressure” and “stable flow” at each stack position before applying stack current.
4. Apply DC current to the stack(s) only once hydraulics are stable, and confirm conductivity in the target stream stabilizes at the expected value.
5. Bring additional stacks online one at a time if the skid has more than one — the log notes that current draw on an already-running stack visibly drops as a newly started stack begins drawing current, which is expected stack-to-stack interaction, not a fault.

5.4 Setting the operating parameters

Dosing pumps are controlled via a 4–20 mA analog signal from ADAM I/O modules. Default configuration: **SET 1** = 4.0 mA (minimum, pump stopped, 0 impulses/min), **SET 2** = 20.0 mA (maximum, 180 impulses/min); internally, a raw integer of 0 corresponds to 3.8 mA and 4095 to 19.7 mA. Each dosing channel is verified across 4/8/12/16/20 mA. TODO: hydraulic setpoints (target flow/pressure) and stack-voltage setpoints are referenced in HydroVolta's internal user-manual outline but no actual values were found — see also appendix A.

5.5 Functional test

A real, filled-in functional test record exists for the BASF pilot's pretreatment dosing pumps and control panel:

1. Verify each dosing channel's SET 1/SET 2 defaults (4.0 mA / 0 impulses/min, 20.0 mA / 180 impulses/min) and confirm out-of-range signals (below 4 mA, above 20 mA) trigger a stop.
2. Test every alarm digital input and dosing digital output for correct read/operate behavior.
3. Drive each dosing pump across 4/8/12/16/20 mA and confirm the correct response at each step.

See the relevant Part II chapters for the treatment (SonixED™/stack) side, pumps, and sensors.

5.6 Handover

Leave a copy of this manual and the unit's quality report on site with the customer.

CHAPTER

Daily operation

IN THIS CHAPTER

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NOTE

This chapter now draws on HydroVolta's own *HMI Description* specification (revision V5, Morocco project, dated 2026-06-16) — the design document the automation team is building the PLC program against. It describes screens, controls and workflow *as specified*; the PLC program itself was still in development at the time of writing (chapter 1), so exact on-screen wording/layout on a specific delivered unit may differ slightly from what is described here. Where an older pilot's tag names differ from this specification (e.g. TSA-prefixed tags used elsewhere in this manual), both are noted.

6.1 Operating elements

The local control panel exposes physical Start, Stop and Reset controls, a Manual/local selector, a feedwater-source selector, and a CIP-mode input (control-system tags `Start`, `Stop`, `Reset`, `Manual`, `FW_Select`, `LS_Select`, `CIP`), alongside a four-color status LED stack (`GREEN_LED`, `BLUE_LED`, `ORANGE_LED`, `RED_LED`). The HMI specification additionally confirms a **hardware Emergency Stop button that bypasses the PLC entirely** — see section 6.9.



Figure 6.1. Suggested figure: photograph of the local control panel with each button, selector switch and LED labeled and cross-referenced to the tag names above.

6.2 Plant modes

Per the HMI specification, the unit operates in one of nine defined modes, selected from the **Mode Control** screen (section 6.4):

1. **Initial fill** — automatic filling of the system before first production.
2. **Production (forward)** — normal automatic production.
3. *(reversal is a timed sub-state of production, not a separate numbered mode — see Starting the unit below)*
4. **Controlled shutdown** — an orderly stop sequence.
5. **CIP** — automatic clean-in-place, after a CIP-PREP step (High pH/NaOH or Low pH/HCl, chapter 12) has conditioned the loop.
6. **Flush/rinse** — freshwater flush of the process loop.
7. *(reserved)*
8. **Manual** — individual device control, access-gated (see *Manual mode* below).
9. *(reserved)*

A separate **Safe Stop** state exists outside the numbered mode list: it is entered automatically on certain faults and requires the recovery procedure in *Emergency stop and restart* below before the operator can select a new mode.

6.3 Starting the unit

Per the HMI specification, starting the unit is a two-step mode selection from the **Mode Control** screen: first **Start AUTO INITIAL FILL**, then, once initial fill completes, **Start AUTO PRODUCTION**. Production runs in *forward* flow with a periodic electro dialysis- reversal step (chapter 10); the Overview screen shows a countdown timer to the next reversal. TODO: the specification confirms the button sequence and mode names but does not itself state numeric pre-start interlocks/permissions (e.g. minimum tank levels) — those are defined in the PLC program, still in development.

6.4 HMI screens

Per the HMI specification, the operator interface is organized into six screens:

- **Overview** — current plant mode and Safe Stop indicator; key performance indicators (product conductivity, product flow, recovery); product routing (to customer vs. recirculation); a status badge for each stack (power on/off, current, conductivity, pressure-OK); pump run/fault status; the position of the critical reversal and CIP-loop valves; the reversal countdown timer; and the active alarm count with highest severity.
- **Tank Summary** — level and conductivity for every tank, and the state of every fill/drain valve.
- **Mode Control** — the plant-mode buttons described above, plus *Drain Concentrate Tank*, *Switch to Manual*, *ACK Alarm* and *Reset Trip*.
- **Alarm Summary** — active, acknowledged and historical alarms, each showing its code, severity, cause, effect, timestamp and a recommended action, following the ISA-18.2 alarm-management convention — see *Alarm handling* in chapter 7.
- **Maintenance / Manual** — individual device commands (pumps, valves, dosing pumps, stack power), gated by the access level in *Manual mode* below.
- **Trends** — historical KPI graphs.

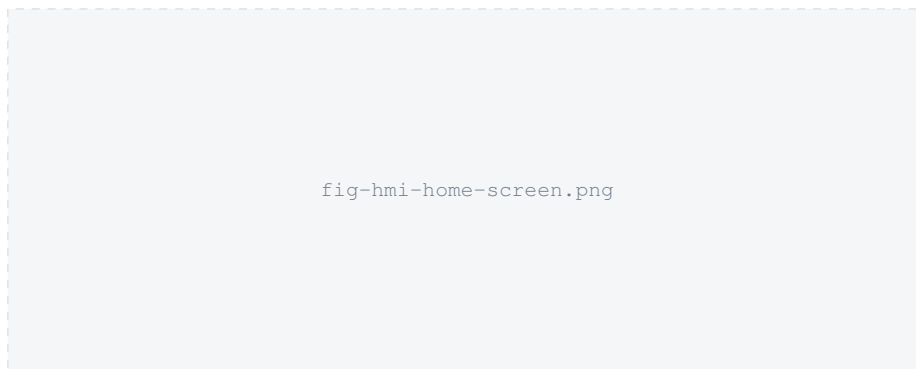


Figure 6.2. Suggested figure: screenshot of the HMI Overview screen, once the PLC program is complete and running on an actual unit, annotated with each element's function per the list above.

6.5 Reading the process values

Confirmed process readings available to the operator include pH, pressure, flow and conductivity/level for the sensor array (TSA_pH, TSA_pressure, TSA_Flow, TSA_Cond, TSA_Level), plus conductivity and level for the TB tank (TB_Cond, TB_Level) and level for the TE (electrolyte) tank (TE_Level) — see chapters 13, 15, 16 and 18. Additional electrical metering is available via power-meter tags (PM3150).

NOTE

HydroVolta's I/O tag lists follow a readable naming convention that helps when reading raw tag names on an HMI or historian trend: a leading letter indicates the measurement type (P = pressure, FL = flow), followed by c/d/e for concentrate/diluate/electrolyte, and S1–S4 for the stack number when the tag is per-stack. Tank tags use T plus a capital letter for the tank's role (TD = diluate/feed tank, TC = concentrate loop tank, TE = electrolyte tank, TB = product/breaking tank). Valve tags use VA plus a functional group number.

6.6 Changing setpoints

Per the HMI specification, every setpoint, threshold, timer, dose and recipe value is adjustable directly from the HMI at runtime, without re-downloading the PLC program. Which parameters a given user can change is governed by a three-level access hierarchy:

- **Operator:** monitor screens, acknowledge alarms, start/stop production, drain the concentrate tank, run a manual flush, view trends. Cannot change setpoints.
- **Maintenance:** everything an Operator can do, plus individual pump/valve/dosing-pump control, manual fills/drains, and non-recipe parameter changes.
- **Admin/Engineer:** everything Maintenance can do, plus stack-power control (as a set, not individually), recipe parameters, cloud command flags, and user-account management.

TODO: the specific PIN/login mechanism for each access level, and the numeric default/range for every adjustable parameter, are defined in the full HMI specification's parameter table but were not transcribed into this manual — consult HydroVolta engineering for the complete parameter table if commissioning a unit before this manual's next revision.

6.7 Manual mode

A `Manual` digital input and `LS_Select/FW_Select` mode-selection inputs exist in the control I/O map, and the HMI specification confirms **Manual** as Mode 8, reached via *Switch to Manual* on the Mode Control screen and gated to the Maintenance/Admin access levels above — an Operator-level user cannot enter Manual mode. In Manual mode, the Maintenance/Manual screen exposes direct command of every process pump, every 2-way and 3-way valve, all five dosing pumps, and stack power (as a set only, even in Manual). One routing valve (`VAR.S.5` in the specification's tag scheme) is explicitly **AUTO-only** and cannot be operated from Manual mode under any access level — do not attempt to command it manually.

6.8 Stopping the unit

A `Stop` digital input exists on the local control panel, and the HMI specification's **Controlled Shutdown** (Mode 4) is the normal, orderly way to stop production from the Mode Control screen — prefer it over simply removing power, so the system can close valves and stop pumps in a defined sequence rather than an uncontrolled one.

6.9 Emergency stop and restart

The HMI specification confirms a **hardware Emergency Stop button that bypasses the PLC entirely** — pressing it removes power/motion directly, independent of software state. TODO: its physical location on the control panel/cabinet was not confirmed in the documents reviewed; verify on the actual unit and update fig. 6.1 accordingly. Certain faults also trip the system into a software **Safe Stop** state (distinct from the hardware E-stop) — both require the following recovery sequence before production can resume, per the HMI specification:

1. Investigate the cause using the Alarm Summary screen (section 6.4) — each entry shows its cause, effect and a recommended action.
2. Physically inspect and resolve the underlying condition.
3. Press **ACK Alarm** on the Mode Control screen.
4. Press **Reset Trip** — this button is only enabled once the fault condition has actually cleared.
5. Select the next mode (e.g. Production). The PLC re-validates all permissives before allowing the mode to start, and will reject the request with a message if a required condition is still not met.



Never attempt to defeat, bridge, or otherwise bypass the hardware Emergency Stop button. If the E-stop has been pressed, do not restart the unit until the cause has been fully investigated and resolved.

Maintenance and troubleshooting overview

General schedule and fault-finding approach — for a specific compartment’s checks, specifications and fault table, see its own chapter in Part II.

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7.1 Safety during maintenance

Before performing any wiring or sensor checks, isolate and verify the control circuit: control voltage is referenced to ground at 24 V DC — confirm 24 V DC is present between the terminal and PE before assuming a wiring fault, and never connect a sensor to the PLC before bench-testing it (chapter 13). TODO: no dedicated safety document (lockout/tagout, PPE, chemical/electrode handling) specific to SalinBloc™ maintenance was found — see chapter 2.

7.2 Maintenance schedule

No complete maintenance interval table (daily/weekly/monthly/annual) was found for the SalinBloc™ system as a whole. The one concrete, recurring interval documented anywhere in HydroVolta's engineering library is the CIP (clean-in-place) cycle described in chapter 12, which ran approximately **weekly** in the documented pilot; the pretreatment UF stage is otherwise cleaned on a TMP-triggered basis rather than a fixed interval (chapter 8).



NOTICE

The table below is a **recommended starting schedule**, not a HydroVolta-issued one. The two rows marked "Confirmed" come directly from source documents cited elsewhere in this manual; every other row is this manual's synthesis of general good practice for equipment of this type (visual/level checks, periodic sensor verification, periodic actuator exercise, annual inspection), pending review and correction by HydroVolta engineering once real field data exists. Do not treat the unconfirmed rows as validated intervals.

Table 7.1. Recommended starting maintenance schedule

Task	Interval	Source / basis
Visual walk-round: check for leaks, unusual noise/vibration, tank levels, HMI alarm state	Daily	Recommended practice (unconfirmed)
Record process values in maintenance log (appendix A ranges)	Daily	Recommended practice (unconfirmed)
CIP cycle (acid + hypochlorite steps, chapter 12)	Weekly	Confirmed — documented pilot interval
Check dosing chemical tank levels and top up	Weekly	Recommended practice (unconfirmed)
UF chemically enhanced backwash (CEB)	When TMP rises ≥ 0.9 bar above initial	Confirmed — UF manufacturer spec (chapter 8)
Bench-verify each 4–20 mA sensor loop (section 13.4)	Quarterly, or on any suspect reading	Recommended practice (unconfirmed)
Exercise/inspect valve actuators (manual override, seat wear)	Quarterly	Recommended practice (unconfirmed)
Inspect control cabinet (dust ingress on IP55 enclosure, terminal torque)	Semi-annually	Recommended practice (unconfirmed)
Full system inspection and log review	Annually	Recommended practice (unconfirmed)

7.3 Maintenance log

A maintenance log should record, per event: date, technician, equipment/asset ID (e.g. stack, pump, sensor tag), the check or service performed, readings taken (pressure, flow, pH, conductivity, etc.) versus expected range, parts replaced (with part number), and any follow-up action required. Its purpose is a traceable per-unit service history so recurring faults, consumable replacement intervals and warranty-relevant events can be reviewed over the system’s lifetime.

Table 7.3. Maintenance log template

Date	Technician	Equipment / tag	Check / service performed	Result

7.4 Alarm handling

Per HydroVolta's HMI design specification (chapter 6), alarms follow the ISA-18.2 alarm-management convention and always resolve the same way, regardless of which alarm fired:

1. The PLC raises the alarm; it appears on the **Alarm Summary** screen with its code, severity, cause, effect, timestamp and a recommended action.
2. The operator investigates using that information and this manual's Troubleshooting section for the relevant compartment (below).
3. Once the operator has acknowledged they are aware of it, press **ACK Alarm** (Mode Control screen) — the alarm moves to the acknowledged list.
4. Physically resolve the underlying condition.
5. Press **Reset Trip** — only enabled once the condition has actually cleared, so a reset that won't take is itself informative: it means the underlying condition has not actually gone away yet.
6. Resolved alarms remain in the alarm *history* with their full timestamp trail, for the maintenance log (chapter 7).

A fault severe enough to trip **Safe Stop** needs the fuller recovery sequence in section 6.9, not just ACK + Reset.

7.5 How to use the troubleshooting sections

Each Part II chapter ends with a *Troubleshooting* section scoped to that compartment: symptom, likely cause, and remedy. Start with the compartment most closely tied to the symptom (e.g. a conductivity alarm → chapter 13 or the relevant tank chapter (chapters 15 to 18); a communication timeout → chapter 14 or chapter 19). If the fault spans several compartments or persists after following the relevant section, escalate per section 7.6.

7.6 When to call service

Escalate to HydroVolta service when: a sensor fails its bench test (chapter 13) and no spare is available on site; a remote-access device (e.g. the ADAM-3600 RTU) remains unreachable after the documented recovery steps (chapter 14) and is not even pingable; or process readings remain outside the documented operating envelope after checking feedwater quality, pump duty point and pipe sizing. See chapter 1 (Contact and support) for contact details.

CHAPTER

Pretreatment and Filtration Skid

Optional add-on — not part of the standard SalinBloc™ unit

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 **NOTICE**

This chapter only applies if the optional pretreatment container was ordered alongside the unit. The standard SalinBloc™ product is a single container with no pretreatment stage — feed water must already meet the stack's feedwater envelope (chapter 10) on its own. If no pretreatment container was supplied, skip this chapter entirely.

8.1 Overview and function

Where fitted, the pretreatment container conditions raw feed water before it reaches the SonixED™ stacks: disc filtration removes coarse suspended solids, and an ultrafiltration (UF) stage removes fine particulates and reduces fouling potential. Filtered water passes to the main unit (chapter 3); rejected solids and backwash water are routed to the drain tank.



Figure 8.1. Suggested figure: photograph of the assembled pretreatment skid inside the container, with the disc filters, UF unit and main piping runs labeled.

8.2 Key specifications

- **Ultrafiltration:** Akualys/AKKIM APM 80 hollow-fiber UF module(s), 2" Victaulic connections. Two modules are confirmed ordered for HydroVolta (160 m² total membrane area). Membrane: modified PVDF hollow fiber, outside-to-inside flow, fiber OD/ID 1.4/0.8 mm, nominal MWCO $\leq 150,000$ Da (≈ 30 nm pore size), 80 m² active area per module.
- **UF feedwater limits:** temperature ≤ 40 °C (typ. 25 °C); turbidity ≤ 250 NTU (typ. 50 NTU); TSS ≤ 80 mg/l; TOC ≤ 30 mg/l; pH 6–9; instantaneous free chlorine ≤ 150 mg/l.
- **UF operation:** dead-end/crossflow, filtrate flux 45–180 l/m²h at 25 °C, flow capacity 3.60–14.40 m³/h per module, feed inlet pressure 2–3 bar (instantaneous max. 5 bar), transmembrane pressure (TMP) 0.4–2 bar, filtrate SDI ≤ 2.5 , filtrate turbidity ≤ 0.2 NTU.
- **Disc filtration:** AZUD Helix Automatic self-cleaning disc filters (FT-series, automatic backwash), 100-micron mesh discs; the documented BASF pilot P&ID additionally references disc-filter positions DF100/DF50/DF20 by micron rating.
- Feed inlet: 2½" (M); filtered-water outlet: 2" (M); drain: 3" (M).
- Feed pump (pilot configuration): submersible, 4.5 m³/h at 6 bar.

8.3 Operation

No dedicated pretreatment-specific operating procedure beyond the general startup/shutdown described in chapter 6 was found, but the UF module manufacturer documents the following normal operating cycle, which is the closest available reference for the standard SalinBloc™ pretreatment skid:

1. **Production:** a brief forward-flush step, then filtration (dead-end or crossflow) until the trigger for backwash is reached.
2. **Backwash:** drain, then air-scour (max. 20 Nm³/h at 600–800 mbar) to loosen the fouling layer, then backwash bottom-to-top and top-to-bottom (backwash pressure max. 2.5 bar) to flush it out.

8.4 Maintenance

The UF manufacturer documents a Chemically Enhanced Backwash (CEB) / CIP cycle, triggered when TMP rises ≥ 0.9 bar above its initial value:

1. Run an extended backwash cycle (as in *Operation* above), then soak the module in cleaning chemical for part of the cycle.
2. CEB chemical concentrations: NaOCl 1000 mg/l, NaOH 500 mg/l, HCl 1000 mg/l, or citric acid 1–2% — select per the fouling type (organic/biological vs. scale) per chapter 12.
3. Total CIP duration ≈ 2 hours, at a cleaning flow of 1–2 m³/h per module.

For extended shutdowns (chapter 20), the manufacturer specifies a glycerol/sodium-metabisulfite preservation soak, and requires a minimum 4-hour flush with ≥ 14 m³ of feed water before restart. Before returning a preserved or newly installed module to service, disinfect with 100 ppm NaOCl circulated for 10 minutes, held for 1 hour, then drained until the filtrate runs chlorine-free. See chapter 7 for the general log template.

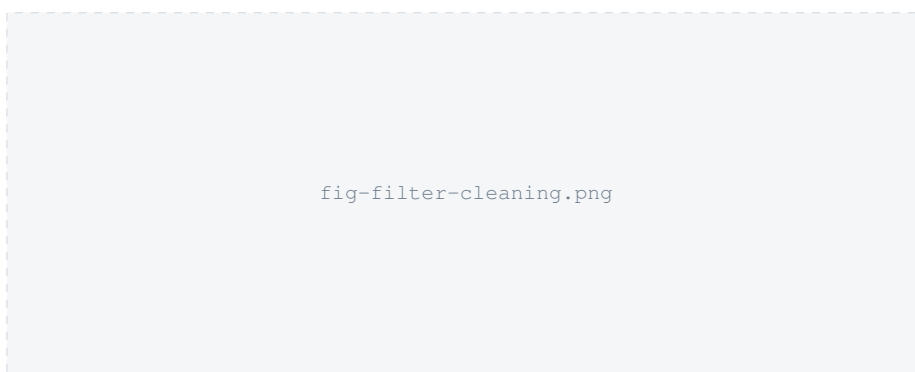


Figure 8.2. Suggested figure: step-by-step photos of the disc-filter backwash and UF CEB/CIP procedure above, once photographed on an actual unit.

8.5 Troubleshooting

- **Low filtered-water flow / high pressure drop:** check pump duty point against the 4.5 m³/h @ 6 bar rating and pipe sizing (60 mm feed line, 40 mm filtered-water line) before suspecting the filters themselves.
- **Rising transmembrane pressure (TMP):** if TMP has risen ≥ 0.9 bar above its initial value, run the CEB/CIP cycle above rather than waiting for a scheduled cleaning.
- **Poor downstream water quality:** verify feed water against the operating envelope in chapter 10 (turbidity, iron, SDI) and against the UF feedwater limits above — an out-of-envelope feed is a likely root cause on either stage.



CHAPTER

Feed and Process Pumps

Feed, diluate and concentrate pumps on each SonixED skid

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9.1 Overview and function

Each SonixED™ skid carries three process pumps — feed, diluate and concentrate — that circulate water through the stack(s) at the required flow and pressure. Pumps are magnetic-driven (sealless) and inverter-(VFD-)controlled so flow can be adjusted without throttling valves.

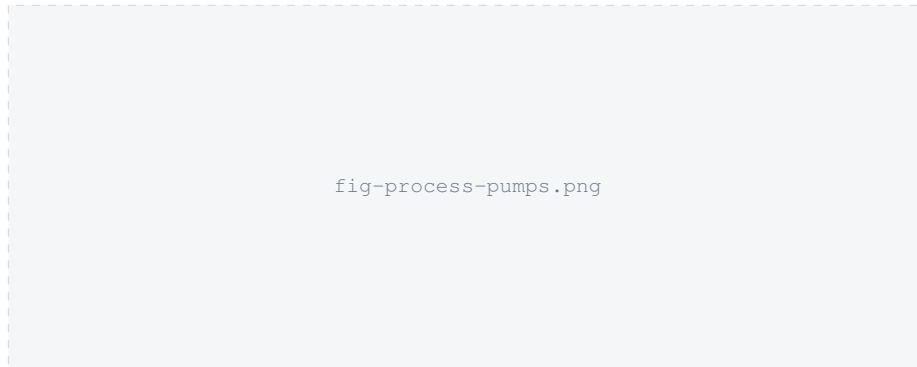


Figure 9.1. Suggested figure: photograph of the three process pumps on a skid, labeled feed/diluate/concentrate, with their VFD units visible.

9.2 Key specifications

- Always 3 pumps per skid (feed, diluate, concentrate).
- Magnetic-driven, inverter-driven, polypropylene/stainless-steel construction, seawater-compatible.
- Pretreatment feed pump (pilot): Grundfos CRN 10-5 A-FGJ-A-E-HQQE, flanged connections.
- Rated hydraulic flow per stack: 24 m³/day (SonixED-S datasheet).
- Each pump is driven by its own VFD (control tags VFD_1_D1...VFD_6_E2 in the pilot configuration).

9.3 Operation

Pump speed is set by the PLC via the VFDs to hold the process at its target flow/pressure. Manual speed override, if available, is reached via the `Manual` mode described in chapter 6.

9.4 Maintenance

TODO: no pump-specific maintenance schedule (bearing/seal inspection, impeller check interval) was found. Magnetic-driven pumps have no shaft seal to replace, which reduces routine maintenance versus a conventional centrifugal pump, but confirm this against the actual pump model fitted.

9.5 Troubleshooting

Each pump's VFD reports a `FaultWord` that should be checked first on any pump-related alarm. Typical causes to check, in order:

- **Pump not running / VFD fault:** check `FaultWord` on the relevant VFD tag; check incoming supply and control-voltage presence (chapter 14).
- **Low flow / high pressure:** check for blockage upstream (chapter 8) or downstream (stack fouling, chapter 10) before assuming a pump fault.
- **Pump running but no output:** check for air lock (magnetic- driven pumps are not self-priming) — TODO confirm priming procedure.



CHAPTER

SonixED Stack Modules

The electro dialysis stacks at the heart of the unit

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10.1 Overview and function

The SonixED™ stack is the unit's core: an assembly of alternating cation- and anion-exchange membranes, titanium electrodes and spacers, with integrated ultrasound transducers for chemical-free antifouling. Under an applied DC voltage, ions migrate across the membranes from the diluate stream into the concentrate stream, desalinating the diluate. Each skid carries 1–3 stacks (standard configurations 1SS/2SS/3SS).



Figure 10.1. Suggested figure: labeled photograph of a single SonixED™ stack module, showing the electrode housings, membrane stack, ultrasound transducers and 1" BSP connections.



Figure 10.2. Suggested figure: a cutaway/exploded diagram of one cell pair, showing the diluate/concentrate/electrolyte flow paths through alternating membranes — useful for explaining the process to a new operator.

10.2 Key specifications

- Module dimensions (l×b×h): 300 × 400 × 1060 mm; weight ~150 kg; connection 1" BSP.
- Flowrate per stack: 0.5–10 m³/h (skid datasheet); 24 m³/day (stack datasheet).
- Salt rejection: up to 90%; water recovery: up to 98%.
- Operating pressure: 0.5–2.5 bar (typical 0.5–1 bar); maximum 3 bar.
- Temperature: 5–40 °C (stack datasheet) / 5–55 °C (skid datasheet) — TODO reconcile which applies to the finished product.
- DC voltage across stack: 60–600 V; DC current: 3–20 A.
- Cell pairs per stack: 50–250; effective membrane area: 10–50 m² ("long" variant) or 20–100 m² ("square" variant).
- Membrane: Fujifilm Type 10, polyolefin.
- Feed water requirements: salinity ≤90,000 ppm; pH 5–8.5 typical (tolerable 2–13); turbidity <1 NTU; suspended solids <20 µm; SDI(5 min) 10; iron <0.5 mg/l; free chlorine 0.5 ppm continuous / 30 ppm intermittent tolerance.

10.3 Operation

Stacks operate continuously under PLC control once commissioned (chapter 5); electro dialysis *reversal* periodically swaps stream polarity as part of the ultrasound antifouling strategy. Process values (pressure, flow, conductivity per stream) are monitored via the sensors described in chapter 13.

10.4 Maintenance

The membrane is the stack's principal wear item (weight ~150 kg, dimensions and connection as in *Key specifications* above). TODO: part number, reorder code and recommended replacement interval were not found; no CIP or cleaning procedure specific to the SonixED™ stack was found among the documents reviewed.

10.5 Troubleshooting

If salt rejection or recovery drops noticeably below the rated ranges above, check feedwater against the operating envelope in *Key specifications* first — turbidity, iron and free chlorine excursions are the most likely root causes of fouling-related performance loss.



CHAPTER

Valves and Manifold

Automated 3-way valves and the feed distribution manifold

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11.1 Overview and function

Automated 3-way valves direct flow between feed, product and recirculation paths (e.g. for electrodialysis reversal, chapter 10), while the manifold assembly distributes feed water evenly to every stack on a skid.



Figure 11.1. Suggested figure: photograph of the manifold assembly with its 3-way valves, labeled by function (feed, product, recirculation).

11.2 Key specifications

- 3-way automated valves: PVC construction.
- Manifold: DN25–DN63, PVC and stainless-steel fittings.

Two actuated-valve product lines are confirmed present in HydroVolta's component library as candidates for this duty:

- **Georg Fischer (+GF+) e-DIASTAR Type 514** actuated ball valve: PVC-U body, 24 V AC/DC multiturn actuator with integrated open/closed position feedback, built-in heating element and manual emergency override. Available in DN25 (1", PN10) and DN50 (2", PN6) sizes matching this manual's connection sizes elsewhere; seal options EPDM or PTFE/EPDM.
- **Bürkert Type 3003** electric rotary actuator, fitted to quarter-turn ball/butterfly valves: available as on/off or proportional control (accepting/outputting 4–20 mA, 0–20 mA or 0–10 V); standard manual override, adjustable limit switches, integrated heating resistor and torque limiter; UL 94 V0 flame-resistant housing.

Both product lines run on the same 24 V DC control bus documented for the cabinet (chapter 14), consistent with the troubleshooting guidance below.

11.3 Operation

Valves are switched automatically by the PLC as part of the stream- reversal and CIP sequences (chapters 6 and 10). TODO: no document describes manual valve override procedure or fail-safe position on power loss — both candidate actuator families above include a manual override lever, but the fail-safe (last-position vs. defined) behavior on power loss has not been confirmed.

11.4 Maintenance

See chapter 7 for the general schedule and log template.

11.5 Troubleshooting

- **Valve not switching:** check 24 V DC control voltage at the valve terminal against PE before assuming a valve fault (chapter 14).
- **Flow imbalance across stacks:** check manifold for blockage or a partially closed valve before suspecting a pump or stack issue.



CHAPTER

Dosing System

Antiscalant, pH correction and CIP dosing pumps

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12.1 Overview and function

Dosing pumps inject antiscalant and pH-correction chemicals (NaOH, HCl, and in the pilot configuration also H₂SO₄) into the feed stream, and supply CIP (clean-in-place) chemicals for periodic cleaning. The documented pilot configuration used 5 dosing channels.

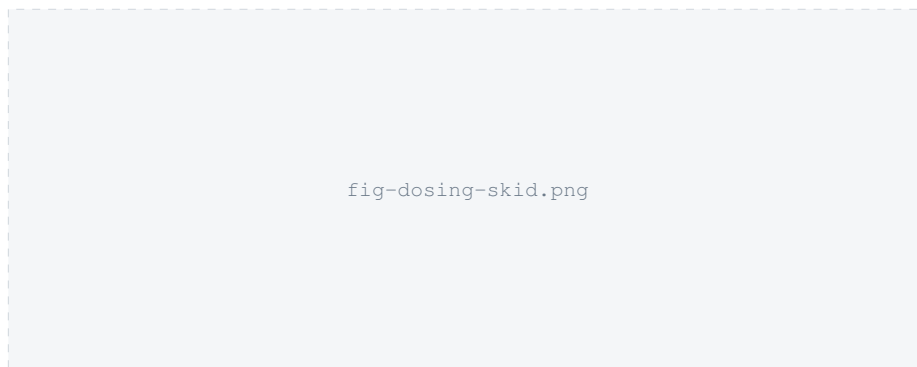


Figure 12.1. Suggested figure: photograph of the dosing skid, with each chemical tank and pump labeled (antiscalant, NaOH, HCl, CIP).

12.2 Key specifications

- 5 dosing channels (pilot configuration), each controlled via a 4–20 mA signal from ADAM I/O modules.
- Signal range: SET 1 = 4.0 mA (0 impulses/min, pump stopped), SET 2 = 20.0 mA (180 impulses/min).
- Chemicals confirmed in the pilot tag list and in Safety Data Sheets held by HydroVolta: antiscalant (AS), NaOH, HCl, NaOCl (sodium hypochlorite) and, in some configurations, H₂SO₄ as an alternative acid.
- Candidate dosing pump: **Etatron GB eOne** series — flow rate 1–30 l/h, max. pressure 20 bar, 100–250 V AC 50/60 Hz supply, max. 300 pulses/min, PVDF pump head with double ceramic ball valves and PTFE diaphragm. The **MF** (multifunction) variant adds proportional dosing, 4–20 mA control, level-switch input and failure alarms; the **PLUS** variant integrates a pH/ORP/Cl controller.

12.3 Operation

See section 5.4 in chapter 5 for the default 4–20 mA signal configuration and functional-test procedure used to verify each dosing channel. Setting a dosing pump's frequency from a 4–20 mA signal uses two settings: the **pulse amplitude** (mL per stroke, set manually at the pump) and the **pulse frequency** (set remotely as a % of the pump's maximum, via the 4–20 mA signal). Worked example from a real HydroVolta dosing calculation (antiscalant, 10 ppm target): dilution factor 1333, pulse amplitude 0.01–0.1 mL, giving a dose rate of ≈ 0.167 mL/min (36 mL/h).



WARNING

Never dose NaOH into feed or process water — HydroVolta's own CIP procedure documents this explicitly as a scaling risk. NaOH is only used against pure (permeate-quality) water during CIP.

A documented weekly CIP cycle (source: HydroVolta pilot CIP procedure) runs as follows:

1. **Acid step:** dose HCl to pH 2.5 in the CIP loop, circulating 30 minutes forward-flow then 30 minutes reverse-flow through the stack, using pretreated feedwater.
2. Drain the CIP loop to the concentrate tank, then flush with fresh feed water for 15–20 minutes.
3. **Hypochlorite step:** dose NaOCl to 2 ppm (effective only near pH 7–7.5 — check/correct pH before this step), circulating 30 minutes forward then 30 minutes reverse.
4. Drain 5 minutes, then flush with fresh feed water for 30 minutes.
5. A variant of this cycle that also treats the electrolyte compartment uses NaOH instead of/alongside HCl — **only ever with pure water**, per the warning above.

Typical HCl/NaOH CIP dose rates from a real pilot's dosing calculation sheet (pure, low-alkalinity water): HCl ≈ 5.5 mL/min at 1000 l/h CIP flow, pulse amplitude 0.1 mL, $\approx 31\%$ of max. pump frequency; NaOH ≈ 17.4 mL/min, pulse amplitude 0.3 mL, $\approx 32\%$ of max. frequency. For natural (higher-alkalinity) feed water, acid demand scales with the water's total alkalinity (TA) — a seawater-range feed (TA 25–30 °f) needs roughly 3–4 $\times$ the acid dose of a low-alkalinity feed. TODO: confirm the CIP interval and dose rates that apply to the standard SalinBloc™ product; the figures above are a real, sourced pilot example, not a general specification.

12.4 Maintenance

TODO: no chemical replenishment interval, tubing/diaphragm replacement schedule, or dosing-pump calibration procedure specific to the standard product was found. The CIP cycle above ran approximately weekly in the documented pilot — use this as a starting reference for the maintenance schedule in chapter 7 until a formal interval is confirmed. Per the chemicals' Safety Data Sheets, store all dosing chemicals in their original, tightly closed containers, away from food/drink, at a recommended **15–25 °C**; keep incompatible chemicals (acids vs. NaOH vs. NaOCl) in separate secondary containment. See section 2.5 for first-aid and spill-response guidance.

12.5 Troubleshooting

Low-level alarms exist per chemical: LL_Alarm_AS (antiscalant), LL_Alarm_NaOH, LL_Alarm_HCl, plus per-channel Dosing1_Alarm...Dosing5_Alarm. On any dosing alarm, check chemical tank level first, then verify the 4–20 mA signal at the pump terminal using the bench-test method in chapter 13. If a CIP cycle fails to reach its target pH (2.5 for the acid step), verify the dosing pump's pulse amplitude/frequency setting against the worked example above before suspecting a pump fault.



CHAPTER

Instrumentation and Sensors

Pressure, flow, pH and conductivity measurement

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13.1 Overview and function

Inline sensors give the PLC (and the operator, via the HMI) a continuous read of the process: pressure, flow, pH and conductivity/level at key points in the pretreatment and treatment trains, plus a pressure gauge on each stack. All sensors use a standard 2-wire, loop-powered 4–20 mA output.



Figure 13.1. Suggested figure: photograph of the inline sensor array on a skid, with each sensor type (pressure, flow, pH, conductivity) labeled.

13.2 Key specifications

- Signal type: 2-wire, loop-powered, 4–20 mA.
- Confirmed sensor types in the pilot tag list: pressure (TSA_pressure), flow (TSA_Flow), pH (TSA_pH), conductivity/level (TSA_Cond, TSA_Level, TB_Cond, TB_Level, TE_Level).
- **Pressure:** WIKA A-10 pressure transmitter (gauge pressure, 4 mA = 0 bar; 2-wire, IP65/IP67, wetted parts 316L stainless steel, non-repeatability $\leq \pm 0.2\%$ of span). Stated maintenance-free by the manufacturer; clean only with water/mild detergent and a soft cloth — repairs are manufacturer-only.
- **Flow:** a real, dated quotation (KROHNE Belgium to HydroVolta, June 2025, explicitly for “meetinstrumenten onziltling” — “desalination measuring instruments”) specifies six **KROHNE OPTIFLUX 1000/1050** electromagnetic flowmeters (DN10–150, compact model, max. operating pressure 16 bar, PFA lining, Hastelloy C22 electrodes, IP66/67, product temperature -25 to $+120$ °C) with an **IFC 050** flow converter.
- **Conductivity:** the same 2025 KROHNE quote specifies **SMARTPAT COND** conductivity sensors.
- **pH/ORP:** the same quote specifies **KROHNE SMARTPAT pH/ORP** sensors — superseding the GF Dry-Loc 2751 candidate noted in earlier pilots.
- **Level:** the same quote specifies an **80 GHz non-contact radar (FMCW)** level sensor (brand consistent with the Holykell/MDS-AM15 candidates found elsewhere, now confirmed as part of a coherent, single-vendor 2025 instrumentation package rather than scattered evaluation quotes).

Because this KROHNE package is dated 2025, covers flow/conductivity/pH/ level in one coherent quotation, and is explicitly scoped to desalination instrumentation (as opposed to the older, scattered GF/Bürkert/Holykell candidates found in earlier pilot projects), it is the **best available evidence for the standard SalinBloc™ instrumentation package** — but it is still a quotation, not a confirmed purchase order or as-built BOM.

- **Optional accessory:** HydroVolta’s SonixED skid datasheet lists a **Cloud Multi-Parameter Water Analyzer (SUP-MPP1000)** as an available option — a 7-inch touchscreen instrument measuring turbidity, pH, conductivity, dissolved oxygen and residual chlorine, with RS485/4–20 mA output, for more detailed product-water quality monitoring than the standard sensor array.

13.3 Operation

Sensor readings feed the process-value displays described in chapter 6 and the alarm limits in appendix A.

13.4 Maintenance

No formal calibration-to-reference procedure (e.g. pH 4/7/10 buffer calibration) was found; what *is* documented is a real bench verification procedure used before commissioning any 2-wire 4–20 mA sensor:

1. Connect the sensor per the electrical schematic up to the terminal strip — **do not** connect to the PLC yet.
2. Insert a test resistor ($\sim 250\ \Omega$, giving a safe 1–5 V signal from the loop) and measure voltage with a voltage-only multimeter (e.g. Fluke 116).
3. For a pressure sensor (4 mA = 0 bar): $\sim 0.88\text{ V}$ across the test resistor confirms a normal 0 bar signal before connecting to the PLC.
4. For a pH sensor (0–14 pH, 4–20 mA) in a 7 pH reference solution: $\sim 2.64\text{ V}$ across a $220\ \Omega$ resistor (12 mA at mid-scale) is expected.
5. If the sensor checks out on the bench but still misbehaves once installed, the fault is in the field wiring, not the sensor.

TODO: this is a wiring/signal-integrity check, not a full calibration procedure — a genuine calibration interval/method for pH, conductivity and pressure sensors was not located.

13.5 Troubleshooting

Table 13.1. 4–20 mA sensor fault table

Symptom (bench test)	Likely cause	Remedy
0 V reading	No control voltage reaching sensor loop	Check DIN connector; confirm 24 V DC between terminal and PE; check polarity; confirm supply is on
Reading below 4 mA	Reversed polarity	Check polarity at the sensor connections
Reading stuck near 4 mA	Sensor connection contamination	Check the connection is dry and clean
Reading above 20 mA	Wiring fault / short	Check connections
No output signal at all (installed)	Cable break	Check loop continuity
Zero point has drifted	Overpressure limit exceeded, or wrong working temperature	Check against rated span and temperature range; recalibrate zero if within spec
Constant output despite pressure/process change	Mechanical fault in the sensor	Replace the sensor
Signal span varies over time	EMC interference	Shield the instrument/cable; remove the interference source
Signal span has dropped	Mechanical overload from an overpressure event	Replace the instrument
Fails on the bench too	Defective sensor	Replace the sensor
Passes bench test, fails installed	Field wiring fault	Inspect and repair wiring in the container

The first four rows above are from the real HydroVolta bench verification procedure; the following five are from the WIKA A-10 OEM manual's own fault-finding section and apply directly to pressure sensors, and by the same 2-wire 4–20 mA principle, are a reasonable first check for the pH and conductivity sensors as well.



CHAPTER

Control Cabinet, PLC and HMI

The IP55 control cabinet, PLC I/O and operator touchscreen

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14.1 Overview and function

The control cabinet houses the PLC, I/O modules, VFDs and terminal wiring that supervise the whole unit: a Weidmüller UC20-WL2000-AC controller (CODESYS/u-OS based) coordinates distributed ADAM I/O modules (ADAM-4024/4050/4017+/5000), the process VFDs (chapter 9), and the dosing subsystem (chapter 12), with an optional touchscreen HMI for local operation.

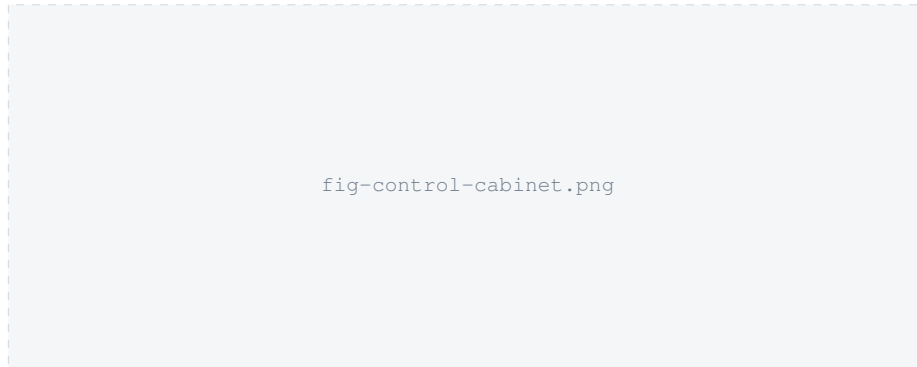


Figure 14.1. Suggested figure: photograph of the open control cabinet with the PLC, I/O modules and terminal strips labeled.

14.2 Key specifications

- Cabinet ingress protection: IP55.
- Controller: Weidmüller UC20-WL2000-AC (CODESYS/u-OS).
- Distributed I/O: Advantech ADAM series. The **ADAM-3600-C2G** Wireless Intelligent RTU (also usable as the main controller) is fully datasheeted in HydroVolta's library: Cortex-A8 AM3352 @600 MHz, 256 MB DDR3L RAM, RT-Linux 3.12; onboard 8AI/8DI/4DO plus a 4-slot I/O expansion bay (compatible modules: ADAM-3613/3617/3618/3624/3651/3656); 1×RS232/485 (DB9) + 2×RS485 (terminal block), 2×RJ-45 10/100 Ethernet; protocols Modbus/TCP, DNP3 L2, IEC 104, MQTT; power 10–30 V DC (24 V @ 5 W typical); operating temperature –40 to 70 °C; DIN-rail or wall mount. ADAM-4024 (analog output, used for dosing pump control), ADAM-4050/4017+ and the ADAM-5000-series backplane (analog I/O cards ADAM-5017P, ADAM-5024) are confirmed in real pilot I/O wiring documentation alongside the ADAM-3600. A more recent (2023) confirmed remote-I/O bill of materials uses **Weidmüller UR20** modular I/O instead: a UR20-FBC-MOD-TCP-V2 Modbus/TCP fieldbus coupler, one UR20-16DI-P (16-channel digital input), one UR20-16DO-P (16-channel digital output), and six UR20-4AI-UI-16 (4-channel universal analog input) modules on a DIN rail — i.e. a Modbus/TCP-based I/O station with analog input as the dominant channel count, consistent with the many 4–20 mA sensors described in chapter 13.
- Control-box ordering options: **CB0** (PLC only), **CB1** (PLC + touchscreen HMI).
- Cloud-connectivity tiers: **CL0** (local only), **CL1** (cloud monitoring), **CL2** (full cloud monitoring and control) — see chapter 19.
- Incoming supply (pilot): 3×400 V AC (3P+N+PE), max. 25 A; 24 V DC control bus.

14.3 Operation

The PLC drives the automatic sequencing described throughout Part II (pump speed, valve switching, dosing) and surfaces process values and alarms to the HMI/dashboard (chapter 6).

14.4 Maintenance

See chapter 7 for the general schedule (dust/filter check on the IP55 enclosure, terminal torque check) and log template.

14.5 Troubleshooting

- **ADAM-3600 timeout / cannot reconnect after uploading a new PLC program:** browse to the device's IP address, log in to its web UI (default password 00000000), go to *Configuration* → *Reboot*. After reboot, retry the stop/connect sequence. Precondition: the device must respond to `ping` — if it does not, the fault is network/ power related, not a program-upload timeout.
- **Modbus-RTU communication issue:** use qModMaster with a CH340 USB-to-Modbus-RTU adapter to verify communication directly: confirm the assigned COM port, set RTU options to match the device (e.g. 9600 baud, 8 data bits, 1 stop bit, no parity), set the correct slave address, use FC1/FC2/FC3, set Base Address to 0, then read a known register to confirm communication. The built-in bus monitor can inspect Tx/Rx traffic for deeper diagnosis.
- **Analog signal fault (any 4–20 mA device wired through this cabinet):** see the bench-test procedure in section 13.4.



CHAPTER

Diluate Tank

Buffers the desalinated stream through recirculation, on its way to the customer

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15.1 Overview and function

The diluate tank (tag prefix **TD**) buffers the desalinated stream as it recirculates through the SonixED™ stack(s) (chapter 10) — diluate is not necessarily delivered in a single pass; it recirculates through this tank and back through the stack until the target product conductivity is reached, at which point it passes on to the product/delivery side of the system (tag prefix **TB**, for “breaking” out of the recirculation loop) for delivery to the customer. This is the tank whose reading is the unit’s headline water- quality result: the HMI Overview screen’s “product conductivity” KPI (chapter 6) is read from this stream.



Figure 15.1. Suggested figure: photograph of the diluate tank(s), labeled with their connections (from-stack, to-stack recirculation, to-product-tank outlet) and instrumentation.

15.2 Key specifications

No tank capacity/material specification exists yet for a finalized standard SalinBloc™ bill of materials, but two real, recent tank orders placed by HydroVolta give a concrete sense of the sizes used: a Rotterdam Plastics order (Oct. 2025) for one 800-litre and one 450-litre vertical polyethylene tank, and a Boralit/Galo Tanks order for two 1000-litre rectangular polyethylene tanks (RT1000RH ZW series). TODO: which of these is the diluate tank was not confirmed in the order documents — do not assume a mapping without checking the as-built P&ID for the specific unit. Some documented configurations use **two** diluate buffer tanks in series (tagged **BTD1/BTD2**, distinct from the main TD tank) rather than one — confirm which arrangement applies to the unit at hand.

- Level: TD_Level (main tank); BTD1/BTD2 level alarms exist in some configurations.
- Process instrumentation on this stream: pH (TD_pH), pressure (TD_pressure), flow (TD_Flow) — see chapter 13 for sensor makes/models.
- Downstream product tank (TB): conductivity (TB_Cond) and level (TB_Level) — this is the KROHNE SMARTPAT COND reading behind the HMI's "product conductivity" KPI (chapters 6 and 13).

15.3 Operation

Diluate recirculates through this tank and the stack under automatic control during **Production** mode (chapter 6); level is monitored continuously with high/low alarms confirmed at **40 cm (low)** / **90 cm (high)** in a documented configuration (see the warning-code table in appendix A) — TODO confirm these setpoints for the standard product's tank sizing. Product only passes to the TB (delivery) tank once conductivity is within its target range.

15.4 Maintenance

TODO: no tank cleaning/inspection schedule was found. Because this tank holds treated (low-salinity) water rather than raw feed or concentrate, the main risk during idle periods is biological growth (algae/biofilm) rather than scaling — inspect at the interval in chapter 7 once one is confirmed.

15.5 Troubleshooting

HL_Alarm_TD/LL_Alarm_TD cover the main tank (plus equivalent alarms for BTD1/BTD2 where fitted). On any level alarm, check the relevant level sensor per the bench-test procedure in section 13.4 before assuming a process fault. If product conductivity (TB) is out of range while the tank level itself is normal, the fault is more likely in the stack's reversal timing or feedwater quality (chapter 10) than in the tank itself.



CHAPTER

Concentrate Tank

Buffers and recirculates the brine/reject stream, and doubles as the CIP reservoir

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16.1 Overview and function

The concentrate tank (tag prefix **TC**) buffers the brine/reject stream as it recirculates through the SonixED™ stack(s) (chapter 10): rejected ions accumulate in this stream over multiple passes, increasing its conductivity, until it is drained for disposal or routed onward for brine valorization (chapter 3). This tank also does double duty as the **circulation reservoir during CIP cleaning** — HydroVolta's documented CIP procedure (chapter 12) explicitly circulates acid and hypochlorite cleaning solution through this same tank, which is why the concentrate tank's minimum level is a precondition for starting a CIP cycle.



Figure 16.1. Suggested figure: photograph of the concentrate tank, labeled with its connections (from-stack, to-stack recirculation, drain/valorization outlet, CIP circulation port).

NOTE

This tank's role as the CIP cleaning reservoir — what water/chemical is in it at each stage, the hazards involved, and how it gets cleaned — is significant and safety-relevant enough to warrant its own chapter: see chapter 17.

16.2 Key specifications

As with the diluate tank, no finalized standard-product capacity/ material specification exists — see chapter 15 for the two real tank orders on file, which are candidates but not confirmed as to role. Some documented configurations use **two** concentrate buffer tanks in series (tagged **BTC1/BTC2**) rather than one.

- **Level:** TC_Level1 (main tank); BTC1/BTC2 level alarms exist in some configurations.
- **Conductivity:** monitored via the “Concentrate Sensor Island” (candidate: KROHNE SMARTPAT COND, chapter 13).
- **Flow:** concentrate-stream flow instrumentation confirmed in the register lists reviewed (chapter 13); a documented alarm exists for concentrate flow below **500 l/hr** (appendix A).

16.3 Operation

Concentrate recirculates through this tank during **Production** mode, increasing in conductivity pass over pass; during **CIP** mode (chapter 12), the same tank instead circulates cleaning chemical. Do not start a CIP cycle without first confirming the concentrate tank's level and that the CIP injection nozzle is submerged (chapter 12).

16.4 Maintenance

TODO: no tank cleaning/inspection schedule was found. Because this tank carries the concentrated reject stream (and periodically, cleaning acid/ hypochlorite), scaling and chemical residue are more likely here than in the diluate tank — inspect and descale per chapter 7 once an interval is confirmed, and always flush with fresh feed water after a CIP cycle per the procedure in chapter 12.

16.5 Troubleshooting

HL_Alarm_TC/LL_Alarm_TC cover the main tank (plus equivalent alarms for BTC1/BTC2 where fitted); a documented alarm fires below 500 l/hr concentrate flow. On any level or flow alarm, check the relevant sensor per section 13.4 before assuming a process fault. If a CIP cycle fails to reach its target pH, verify the concentrate tank's level and injection nozzle first, per chapter 12.



CHAPTER

CIP Tank

The concentrate tank, in CIP mode: what's really inside it, and why that matters

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17.1 Overview and function

No HydroVolta document identifies a separate, dedicated CIP tank — **the concentrate tank (chapter 16) is the one and only vessel used for CIP** cleaning, per HydroVolta's own documented CIP procedure (chapter 12). It gets its own chapter here because what is *inside* that tank changes completely depending on which mode the unit is in, and confusing the two is a real chemical-exposure hazard — read this chapter alongside chapters 12 and 16, not instead of them.



Figure 17.1. Suggested figure: the same concentrate tank as fig. 16.1, with the CIP dosing/circulation connections called out separately from the normal process connections.

17.2 Key specifications: what is in the tank, and when

- **Production mode:** the tank holds **concentrate/ brine** — rejected feedwater ions, increasingly concentrated over multiple passes (chapter 16). This is process water, not a cleaning chemical.
- **CIP acid step:** the tank holds **pretreated feedwater dosed with HCl to pH 2.5** (chapter 12) — an acidic solution, corrosive at that pH.
- **Between CIP steps:** the tank is drained and refushed with **fresh feed water** for 15–20 minutes specifically to clear the acid before the next chemical goes in.
- **CIP hypochlorite step:** the tank holds **feedwater dosed with NaOCl to 2 ppm** (sodium hypochlorite, effective only near pH 7–7.5) — an oxidizing, corrosive solution.
- **NaOH variant of CIP** (electrolyte-compartment cleaning): the tank holds **NaOH dissolved only in pure (permeate-quality) water** — **never** feedwater or process water, per the warning in chapter 12 (mixing NaOH with the minerals present in feed/ process water causes scaling).

No capacity/material specification exists yet for a finalized standard SalinBloc™ bill of materials — see chapters 15 and 16 for the real tank orders on file and their (unconfirmed) role.

17.3 Operation



DANGER

Never let the acid (HCl) and hypochlorite (NaOCl) steps mix in this tank. Mixing an acid with sodium hypochlorite releases chlorine gas, which is toxic even in small quantities. This is precisely why HydroVolta's documented procedure requires a full drain and a 15–20 minute fresh-water flush between the acid step and the hypochlorite step (chapter 12) — never shorten or skip that flush to save time, and never dose both chemicals into the tank at once. If chlorine odor is detected during or after a CIP cycle, stop dosing immediately, ventilate the area, and do not re-enter until it has cleared — see section 2.5 for first-aid measures for chemical/inhalation exposure.

Both the acid and hypochlorite solutions are corrosive at the concentrations used (see the Safety Data Sheet summary in section 2.5): treat the tank, its vent, and any sample point as a chemical-exposure risk during CIP, not just during dosing-pump refilling. Wear the PPE specified in section 2.3 whenever the tank is opened, sampled, or drained during or immediately after a CIP cycle. Do not start a CIP cycle without first confirming the tank's level and that the CIP injection nozzle is submerged (chapter 12).

17.4 Maintenance: cleaning the tank itself

The documented CIP procedure already *is* the tank-cleaning procedure — there is no additional physical scrub-down step on file. In order: drain the acid step, flush with fresh feed water 15–20 minutes; drain the hypochlorite step 5 minutes, then flush with fresh feed water 30 minutes (chapter 12). TODO: no document describes a physical inspection (scale/residue check on the tank walls, fittings, level sensor) as part of or after this cycle — until one exists, a visual check after the final flush (before returning to Production mode) is good practice. See chapter 16 for the tank's general (non-CIP) maintenance notes.

17.5 Troubleshooting

If a CIP cycle fails to reach its target pH (2.5 for the acid step), verify the tank's level and injection nozzle before suspecting a dosing- pump fault (chapter 12). If chlorine odor is ever detected, treat it as an emergency per the warning above, not a troubleshooting item to work through calmly. `HL_Alarm_TC/LL_Alarm_TC` (chapter 16) apply during CIP exactly as they do during Production — do not disable or ignore them because the tank is in CIP mode.

Electrolyte Tank

A separate loop feeding the stack’s electrode compartments — never part of the treated-water path

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18.1 Overview and function

The electrolyte tank (tag prefix **TE**) supplies a separate, recirculating loop that feeds the SonixED™ stack's **electrode compartments** (chapter 10) — this is what actually carries current at the electrodes. Unlike the diluate and concentrate tanks above, the electrolyte loop is **not part of the treated-water path**: it never mixes with the feed, diluate or concentrate streams, and is not delivered to the customer. A documented “Electrolyte Preparation” monitoring point (its own conductivity transmitter, distinct from the main process conductivity sensors) confirms the electrolyte solution is prepared and monitored as its own loop, separate from process water quality.



WARNING

Electrode reactions can evolve gases. The system carries a general “Ventilation OFF” alarm (documented warning code W002, appendix A) — while that alarm is a general system warning rather than confirmed as electrolyte-specific, adequate ventilation matters most wherever electrolysis is taking place, i.e. around the stack and electrolyte loop. TODO: HydroVolta engineering should confirm the electrolyte chemistry and any specific gas-hazard/ventilation requirement for this loop — see also chapter 2.

fig-electrolyte-tank.png

Figure 18.1. Suggested figure: photograph of the electrolyte tank and its connections to the stack's electrode compartments, clearly distinguished from the diluate/concentrate piping.

18.2 Key specifications

No finalized standard-product capacity/material specification exists — see chapter 15 for the two real tank orders on file (candidates, role unconfirmed). A documented level-alarm configuration uses narrower thresholds for this tank than for diluate/concentrate — **20 cm (low) / 80 cm (high)** (appendix A) — consistent with this being a smaller support loop rather than a main process buffer, though this has not been confirmed as a general rule.

- Level: TE_Level.
- Electrolyte-preparation conductivity: monitored separately from the main process streams (chapter 13).

TODO: electrolyte solution composition (which chemical, target concentration) and replenishment/dosing method were not found in HydroVolta's documentation reviewed.

18.3 Operation

The electrolyte loop recirculates continuously whenever the stack is powered (chapter 10); level is monitored via `TE_Level` with the narrower alarm band noted above.

18.4 Maintenance

TODO: no maintenance schedule for the electrolyte loop was found. Because this loop directly feeds the electrode compartments, its condition is tied to electrode/membrane health — see the stack maintenance notes in chapter 10 for the (also largely undocumented) membrane/electrode service interval.

18.5 Troubleshooting

HL_Alarm_TE/LL_Alarm_TE cover this tank. An unexpected electrolyte level drop should prompt a check for leaks at the electrode compartment seals (chapter 10) before assuming a sensor fault; verify the sensor itself per section 13.4 either way.



CHAPTER

Telemetry and Remote Connectivity

Router, remote-access gateway and cloud monitoring

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19.1 Overview and function

A cellular router and cloud dashboard give operators and HydroVolta service remote visibility into the unit's process values and alarms, independent of the local HMI (chapter 14).



Figure 19.1. Suggested figure: photograph of the router/gateway mounted in the cabinet, with antenna and SIM slot visible.

19.2 Key specifications

- **Confirmed deployed remote access:** real OpenVPN client configuration files exist in HydroVolta's engineering library for at least two pilot containers (Farys, BASF/CAGECE demo), following a router-based “meeting point” VPN architecture (Weidmüller IE-SR-2GT-LAN / IE-SR-2GT-UMTS routers, per the vendor tech note on file) — this is the one telemetry element confirmed as *actually used*, not just evaluated.
- **Confirmed deployed dashboard: ThingsBoard**, with a real MQTT parameter list and full tag list captured from the BASF pilot deployment.
- A real 2023 requirements-workshop record (source: 08_IT/00_OQuila progress) describes an **actually running 4-container pilot fleet** at that time: **Advantech ADAM-3600** connecting directly to **Azure IoT Hub**, piped through **Azure Stream Analytics** into a SQL database and visualized in **Power BI**; connectivity via a **4G Teltonika router**, with HydroVolta noting a possible future switch to Ewon “for a better SLA” due to past Teltonika stability issues. Each container sent a telemetry message roughly every 10 seconds ($\approx 8,000$ messages/container/day: connectivity, pressure, flow, temperature, energy). Parameter changes at that time were only possible via VPN, not from the cloud dashboard.
- **OQuila** was a third-party IoT consultancy HydroVolta held a 2023 requirements workshop with, to evaluate a longer-term telemetry platform (long-term storage, a graphical P&ID front-end, configurable alarm thresholds, role-based access). HydroVolta's own workshop notes state ThingsBoard was assessed as “ok” for the dashboard need on its own. **No document confirms OQuila's proposal was ultimately adopted or built** — treat the Azure IoT Hub/Power BI pipeline above and the ThingsBoard/ Weidmüller-OpenVPN pipeline elsewhere in this manual as two telemetry architectures documented at different points in HydroVolta's history, not necessarily the same one.
- Controller reached over the network: Weidmüller UC20-WL2000-AC (chapter 14).

19.3 Operation

Process values available on the cloud dashboard mirror those on the local HMI (chapter 6).

19.4 Maintenance

See chapter 7 for the general schedule and log template.

19.5 Troubleshooting

- **Router unreachable:** confirm the device is powered and check its cellular signal/SIM status before assuming a software fault.
- **Cloud dashboard shows stale data:** check the router's connection to the controller first; if the controller itself is reachable locally but not remotely, the fault is likely in the router/network path rather than the controller.

See also the ADAM-3600 and Modbus-RTU troubleshooting steps in chapter 14, which apply to field-device connectivity issues more broadly.



CHAPTER

Shutdown, storage and disposal

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20.1 Short shutdown

Use **Controlled Shutdown** (Mode 4) from the Mode Control HMI screen (chapter 6) rather than simply removing power, so pumps and valves stop in a defined sequence. If the unit is to be fully powered down (as opposed to left in Controlled Shutdown/standby), isolate at the incoming-supply breakers (main isolator, MCBs, RCDs) per the unit's own electrical schematic (chapter 14).

20.2 Long-term storage

No storage/preservation procedure specific to the whole SalinBloc™ unit was found. Where the optional pretreatment container is fitted (chapter 8), the UF membrane manufacturer's preservation procedure is the best-documented reference available for that stage: soak the UF module in a glycerol/sodium-metabisulfite preservation solution before an extended shutdown, and flush with $\geq 14 \text{ m}^3$ of feed water for at least 4 hours before restart. TODO: no equivalent preservation guidance (membrane wetting, antifouling flush, dosing-system winterization) was found for the SonixED™ stack itself or the dosing system — for the dosing system, at minimum follow the chemical storage guidance in chapter 12 (15–25 °C, tightly closed original containers) for the duration of the shutdown.

20.3 Restarting after storage

Treat a restart after storage as a fresh commissioning (chapter 5).

20.4 Disposal and recycling

Component datasheets (e.g. dosing pumps, chapter 12) may carry manufacturer disposal notes; dispose of dosing chemicals per their Safety Data Sheets (section 2.4) and local regulations.

Parameter tables

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A.1 Default setpoints

Table A.1. System performance defaults (SonixED skid v2025-A7)

Parameter	Value
Flowrate per stack	0.5–10 m ³ /h
System flow per skid	up to 30 m ³ /h
Recovery rate	up to 98%
Operating pressure	0.5–2.5 bar
Temperature range	5–55 °C
Specific energy	0.6– 2.0 kWh/m ³
Skid dimensions (l×b×h)	3040×1340×2020 mm
Incoming line voltage	3×400 V AC
Max. current	25 A
Valve voltage	24 V DC
Control voltage (DC bus)	24 V DC
Dosing SET 1 (min)	4.0 mA / 0 impulses/min
Dosing SET 2 (max)	20.0 mA / 180 impulses/min

Table A.3. SalinBloc™ container throughput by format (source: HydroVolta brochure)

Parameter	20 ft unit	40 ft unit
Flow rate	2–30 m ³ /h	15–60 m ³ /h
Approx. daily throughput (22 h/day operation)	44– 660 m ³ /day	330– 1,320 m ³ /day

Feed range for both formats: brackish groundwater, 1,500–20,000 ppm TDS. TODO: no container weight figure was found in any source reviewed, including the container general-arrangement drawings themselves (their title blocks carry no weight/dimension text field, and any dimension lines are vector geometry, not extractable text — a human must open these drawings visually). Obtain the weight from the container manufacturer or a loaded-unit weighbridge reading before publishing lifting/foundation guidance.

A.2 Site-specific settings

Use this table to record the as-commissioned values for this specific installation.

Table A.5. Site-specific settings template

Parameter	Value

A.3 Alarm limits

The P&ID (appendix B) shows instrument spans used as default alarm spans at each skid position: pressure 0–2.5 bar, conductivity 0–200 mS/cm, with high/low alarm tags (PSA, CEA) at every position. HydroVolta's ThingsBoard warning-code table (a real, deployed alarm definition list, source: 08_IT/01_Thingsboard/Warning table) gives concrete numeric examples for several alarm classes:

Table A.7. Representative alarm/warning codes (source: HydroVolta ThingsBoard warning table)

Code	Class	Condition
W001	General	HV-Skid not powered
W002	General	Ventilation off
W011–W013	Chemical	Antiscalant / H ₂ SO ₄ / HCl tank empty
W101–W110	Level	Main diluate/concentrate/electrolyte/breaking/client tank level low or high (e.g. diluate below 40 cm or above 90 cm; electrolyte below 20 cm or above 80 cm)
W111–W118	Level	Buffer tank low-low / high level
W201–W214	Electrical	Stack stage 1–7 voltage below or above system specification
W300	Flow	Diluate flow below 500 l/hr
W301	Flow	Concentrate flow below 500 l/hr

TODO: this table only samples the documented code ranges; the full warning table has ~57 defined codes and should be transcribed in full by HydroVolta engineering, together with the exact level/pressure setpoints per tank size, once the standard SalinBloc™ tank configuration (chapters 15, 16 and 18) is finalized — these codes come from a specific pilot project's tank/stage configuration and the numeric thresholds may not transfer directly. See the alarm tag list in chapters 6, 12, 15, 16 and 18 for additional alarms.

APPENDIX

Drawings

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B.1 P&ID

Piping & instrumentation diagram for the containerized BASF pilot water treatment system: drawing no. SWHC-001/00, rev. 0.98, dated 11-11-2022, “BASF WATER Container” (Grant Agreement 969116 — SonixED, H2020-EIC-SMEInst). Shows pretreatment filtration skids (SP-01–SP-4), dosing units for antiscalant/NaOCl/NaOH/HCl/H₂SO₄ (CIP), tanks, and instrumentation with alarm tags across the treatment train.



Figure B.1. Insert the full P&ID drawing here (source: SWHC-001/00, rev. 0.98) once confirmed representative of the standard SalinBloc™ product, or the product-specific P&ID once available.

HydroVolta’s standardization library includes an ANSI/ISA-5.1 P&ID symbol legend, which is the reference standard the symbols on this and other project-specific P&IDs (BASF, Saudi, Australia demo, Morocco) are drawn against.

B.2 Electrical schematic

Circuit diagram set “Pretreatment EP” for the pretreatment container, document series C1/I1/I2/E1–E17 (file 31-01-2023 Pretreatment.dsn). Covers the 3-phase incoming supply and breaker/RCD protection (sheet E1), 24 VDC power/control distribution (E2–E3), PLC I/O wiring (E4–E5), terminal wiring (E6–E8), valve/sensor junction boxes (E9–E13), VFD control (E14), sensor/valve wiring standards (E15), dosing pump wiring (E16), and Modbus COM2 wiring (E17).

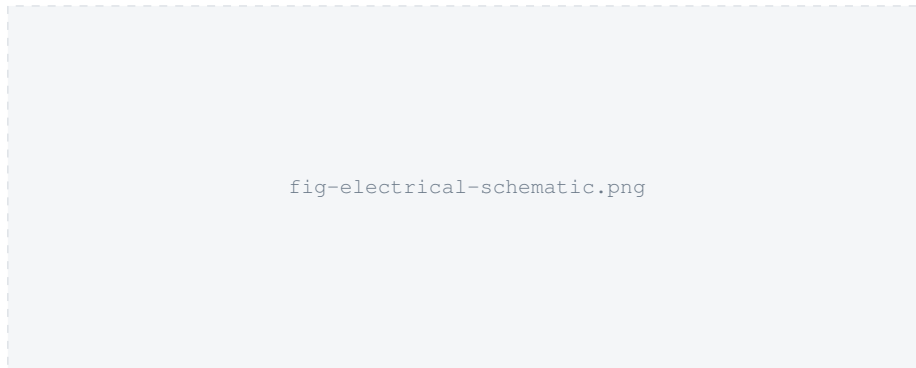


Figure B.2. Insert the relevant electrical schematic sheet(s) here (source: 31-01-2023 Pretreatment.dsn, sheets E1–E17) once confirmed representative of the standard SalinBloc™ product.

B.3 Container layout

20 ft general-arrangement drawing “2023_08_21 SonixED container BASF_2308CA1” (6058 × 2438 × 2896 mm), and companion skid top-view layout “2023_08_22 BASF Skids Top View_2308CA2” — see chapter 3 for the summary description.

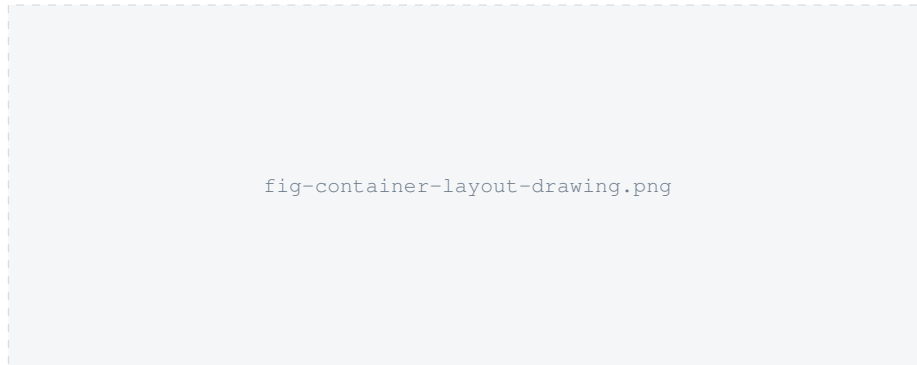


Figure B.3. Insert drawings 2308CA1 (general arrangement) and 2308CA2 (skid top view) here, or the standard-product equivalents once available.

APPENDIX

Certificates and datasheets

IN THIS CHAPTER

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C.1 Declaration of conformity

TODO: only a blank EU Machinery Directive DoC template was found (“EU declaration of conformity-en”, Ref. Ares(2015)1600946), not a signed, product-specific Declaration of Conformity — HydroVolta’s 02_DoC’s CE-marking folder is currently empty. A signed DoC must be inserted here before this manual is issued to a customer.

HydroVolta’s legal/legislation library holds reference texts for the EU directives and regulations most likely to apply to a CE-marked SalinBloc™ unit, which a future DoC should be checked against: the Machinery Directive (2006/42/EC), EMC Directive (2014/30/EU), Low Voltage Directive (2014/35/EU), RoHS, WEEE, the EU Drinking Water Directive (2020/2184) if a potable-water claim is made, and REACH/CLP for the dosing chemicals (chapter 12). A component-level DoC exists for at least one purchased part (the WIKA A-10 pressure sensor, chapter 13) — this is normal for a CE-marked assembly (component conformity feeds the assembly-level DoC) but is not itself a substitute for the unit’s own Declaration of Conformity.

C.2 Component datasheets

The SonixED™ Skid System datasheet (Model SONIXED-SKID, Version 2025-A7) is the primary component reference used throughout Part II of this manual. Individual component-level datasheets (I/O modules, dosing pumps, pumps, pneumatics) exist in HydroVolta's engineering library and should be attached here per shipped configuration.

C.3 Regulatory approvals (drinking water)

No UK DWI Regulation 31 approval and no NSF/ANSI listing (e.g. NSF/ANSI 61 or 372) exists yet for SalinBloc™/SonixED™— HydroVolta's datasheet library holds only generic reference/guidance material on how those approval processes work, not an application or approval record (see section 2.1 in chapter 2). TODO: pursue formal approval before offering the unit for potable-water use in a regulated market, or clearly disclaim potable-water suitability until then.

C.4 Water analysis report

TODO: no formal water-analysis lab report/certificate of analysis was found for the standard product; the closest material found (third-party pretreatment-component water-quality questionnaires from earlier pilot projects) is not a substitute. A per-unit water analysis report should be attached here once available — see also the "Find your unit's quality report" lookup on the online container guide page (chapter 1).

Glossary

Term	Definition
TDS	Total Dissolved Solids — the concentration of dissolved minerals/salts in water.
SonixED™	HydroVolta's electrodialysis-based desalination technology/product line.
ED	Electrodialysis — a membrane process using an electric field to move ions through selective membranes.
EDR	Electrodialysis Reversal — periodically reversing the applied DC field's polarity to clean the membranes in place and stabilize performance.
TMP	Transmembrane Pressure — the pressure difference across a filtration membrane; a rising TMP indicates fouling and triggers cleaning.
CEB	Chemically Enhanced Backwash — a backwash cycle that includes a chemical cleaning soak, used on the ultrafiltration stage.
VCA	Veiligheid, Gezondheid en Milieu Checklist Aannemers — a Benelux contractor health/safety/environment competency certification.
Stack	The assembly of alternating ion-exchange membranes, electrodes and spacers that performs electrodialysis.
Skid	A pre-assembled equipment module (pumps, valves, instrumentation) on a common frame.
Diluate	The desalinated water stream leaving the stack.
Concentrate	The brine/reject stream leaving the stack, carrying the removed ions.
Electrolyte	A separate circulating stream feeding the stack's electrode compartments.
Ion-exchange membrane	A selective membrane that passes either cations or anions, blocking the other.
PLC	Programmable Logic Controller — the industrial controller running the unit's automatic logic.
HMI	Human-Machine Interface — the operator touchscreen/display.
VFD	Variable-Frequency Drive — controls pump/motor speed.
RTU	Remote Terminal Unit — a networked I/O/communication device (e.g. the ADAM-3600).
CIP	Clean-In-Place — an automated cleaning cycle using dosed chemicals.
P&ID	Piping & Instrumentation Diagram.
DoC	Declaration of Conformity — the manufacturer's statement of regulatory compliance (e.g. CE marking).
PPE	Personal Protective Equipment.
E-stop	Emergency stop — a manually operated device that immediately halts machine operation.

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